AGE 316: MICROCOMPUTER SYSTEMS

LECTURE 4: THE ARDUINO IDE & SKETCH STRUCTURE

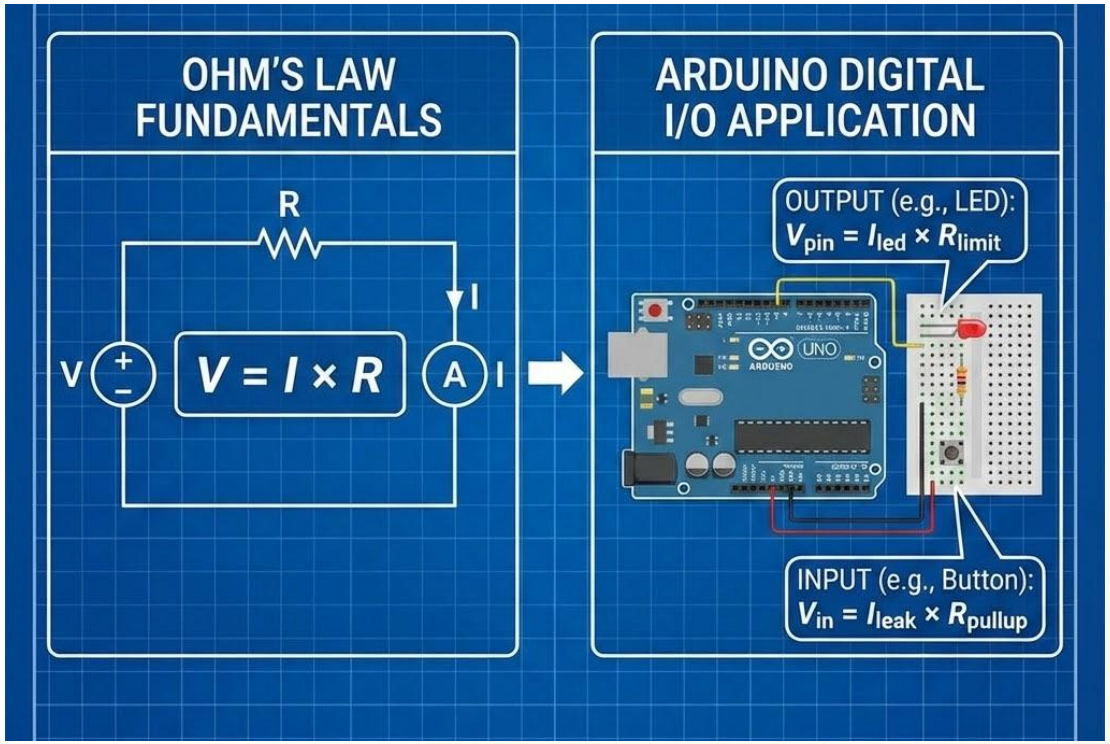
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Recap: Lecture 3

- **Ohm's Law:** V = I × R is the golden rule.
- **Digital I/O:** Pins can be HIGH (5V) or LOW (0V).
- **Functions:** pinMode(), digitalWrite(), digitalRead().

Today's Goal: We move from *hardware theory* to the *software workshop* where we write our logic.



What is an IDE?



Integrated Development Environment



Text Editor

Where you write your code.



Compiler

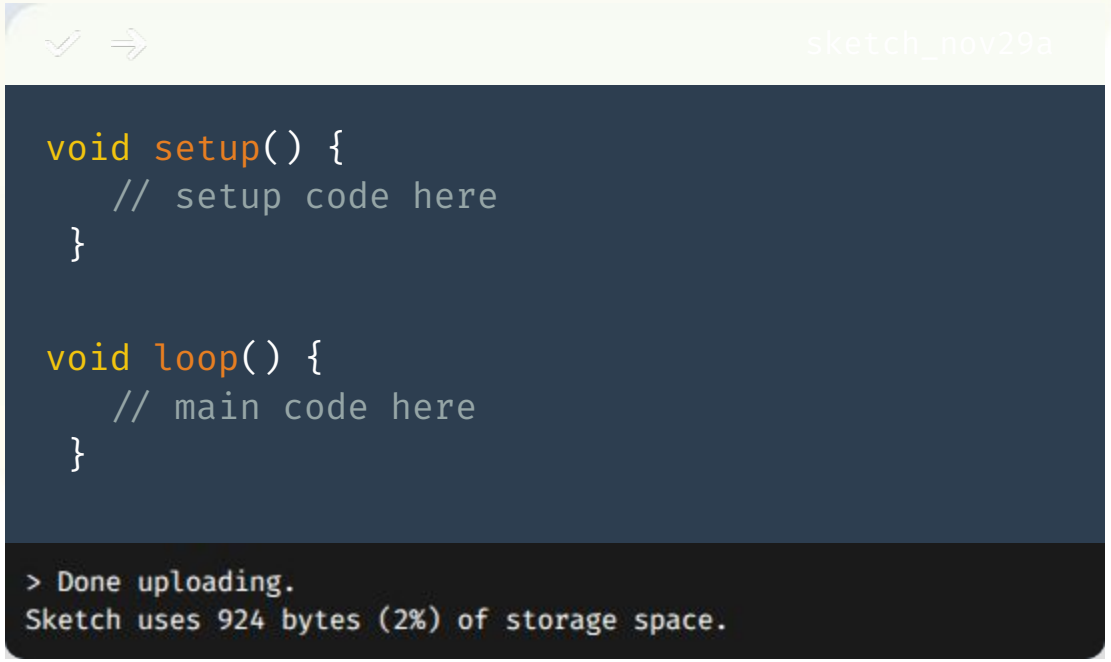
Translates code to machine language (Hex).



Uploader

Sends the binary to the Arduino board.

The Arduino Interface



- **Verify (Check):** Checks your code for syntax errors.
- **Upload:** Compiles and sends code to the board.
- **Console:** Shows status and error messages (Black box at bottom).

Terminology: The "Sketch"



In the Arduino world, a program is called a **"Sketch"**.

This naming comes from the "Processing" language, which Arduino was originally based on, designed for artists and designers.

Anatomy of a Sketch

Every Arduino sketch **MUST** have two specific functions to work:

1. void setup()

Runs **ONCE** when power is applied or reset.

Used for: Initialization, pinMode, starting Serial.

2. void loop()

Runs **REPEATEDLY** forever.

Used for: Main logic, reading sensors, blinking LEDs.

Deep Dive: void setup()

```
void setup() {
   // Prepare pin 13 as output
   pinMode(13, OUTPUT);

   // Start serial comms
   Serial.begin(9600);
 }
```

The Preparation Phase

Think of this as "packing your bag" before a trip.

- You only pack once.
- You define your tools (Inputs/Outputs).
- You set your communication speed.

Deep Dive: void loop()

The Main Engine

Think of this as your "daily routine".

- It happens over and over again.
- It executes top-to-bottom.
- When it reaches the bottom, it jumps back to the top immediately.

🔄 **Infinite Cycle**

```
void loop() {
   digitalWrite(13, HIGH);
   delay(1000);
   digitalWrite(13, LOW);
   delay(1000);
 }
// Jumps back to start!
```

Visual Analogy



SETUP

Waking Up

You only wake up once per day. You put on your clothes and shoes.



LOOP

Daily Tasks

Eat -> Study -> Sleep -> Repeat. You do this continuously until the "power" goes out.

The Grammar of C++

Arduino uses C++, which is strict about punctuation.

```
voi  setu () { // Curly
d pinMod (13, OUTPUT); //
} e           T        Semicolon
```

- **Semicolons (;):** End of a statement. Like a period in a sentence. Missing this is the #1 error!
- **Curly Braces { }:** Define a block of code (start and end).
- **Case Sensitivity:** pinMode is NOT the same as PinMode.

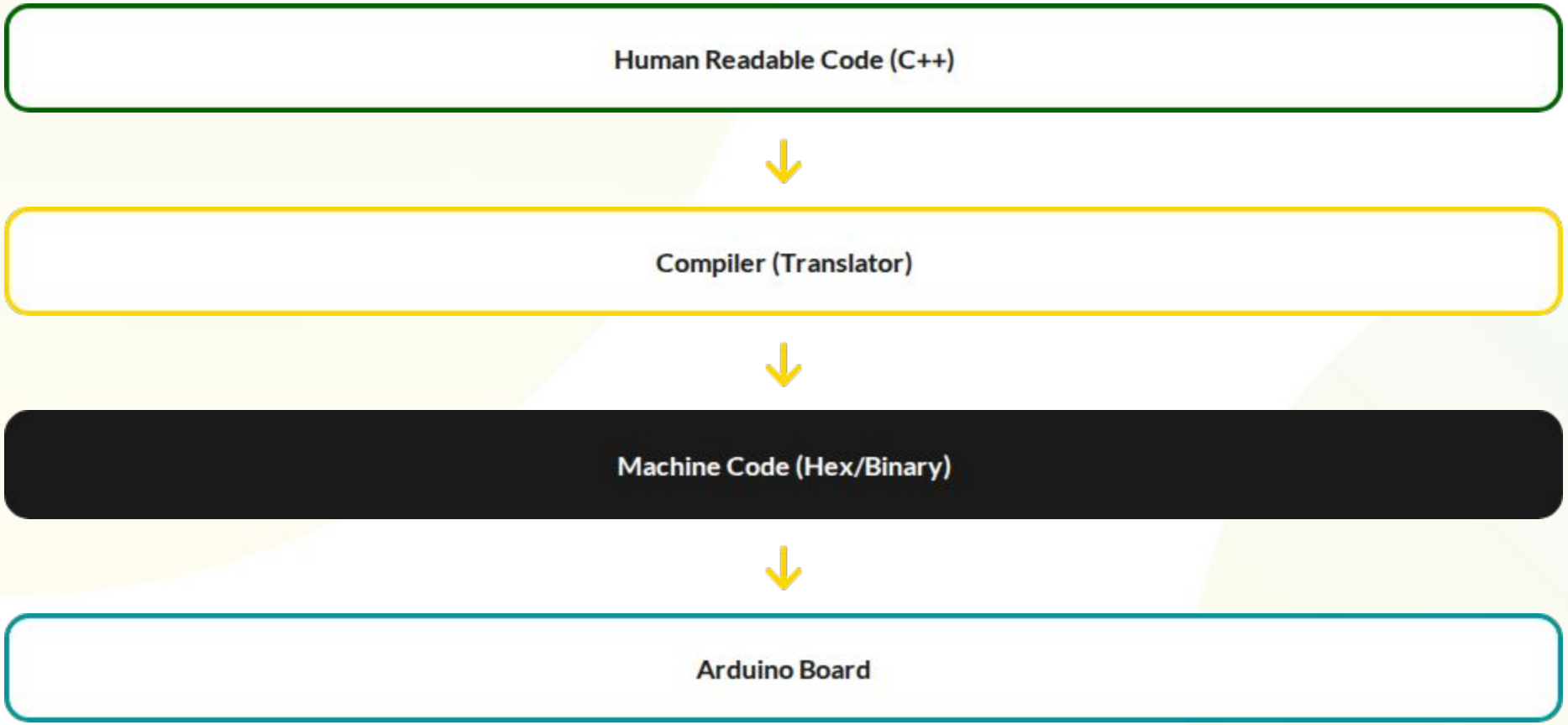
Leaving Notes: Comments

Comments are ignored by the Arduino but helpful for humans.

```
// This is a single line comment
delay(1000); // Wait for 1 second

/*
   This is a multi-line block comment.
   You can write paragraphs here.
 */
```

The Compilation Workflow



Don't Fear the Error!

```
exit status 1
expected ';' before '}' token
```

Common Pitfalls:

1. **Syntax:** Missing semicolon or unmatched curly brace.
2. **Typo:** Writing digitalwrite instead of digitalWrite.
3. **Port Selection:** Forgetting to select the correct COM port in *Tools* > *Port*.

Read the black console at the bottom! It tells you the line number.

Practical: Modifying Blink

Your Task:

1. Open *File > Examples > 01.Basics > Blink*.

2. Upload it and watch the LED.

3. Change the delay(1000) to delay(200).

4. Upload again. What happens?

```
digitalWrite(LED_BUILTIN, HIGH);
delay(1000); // Change this!
digitalWrite(LED_BUILTIN, LOW);
delay(1000);
```

Coding Best Practices

Indentation

Keep your code tidy. Use Tabs to show what is inside the curly braces.

Comments

Write comments so "Future You" knows what the code does.

Naming

Use descriptive names. ledPin is better than x.

Summary

- **IDE:** The workshop where we write and upload code.
- **Setup():** Runs once (Initialization).
- **Loop():** Runs forever (Main Logic).
- **Syntax:** Watch your semicolons and braces!

Questions?

"The only stupid question is the one you don't ask."

Dr. Kadeghe Fue

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